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Introduction

1.0 FOREWORD

This book was written to provide a practical guide to those individuals responsible for providing industrial hygiene (IH) services to the semiconductor industry. The purpose of this book is to help those individuals gain a better understanding of the hazards in the industry and potential approaches to characterizing their evaluation and control, and to share some insights that have been gained by the authors and reviewers.

The intended audience for this book is the practicing industrial hygienist, either new to this industry or embarking on an IH career. The information in this book may also be valuable for facilities engineering personnel, occupational health and safety professionals, and researchers involved in the semiconductor industry.

2.0 OVERVIEW OF BOOK

The book is organized into relevant chapters that use the strategy of looking at the types of hazards (chemical and physical), a specific control technique (ventilation), then direct protection of the individual (personal protective equipment (PPE)/respiratory protection). Because of the emerging importance of indoor air quality, electromagnetic fields, and ergonomics to any comprehensive IH program, these are discussed in detail, as well as approaches to IH record keeping and plan review.

Each of these individual topics could easily be the focus of an entire book. The authors attempt to distill and focus this information into a summary of the most germane, and emphasize the relative significance to safety and health in the semiconductor industry.

3.0 BACKGROUND

As the semiconductor industry has evolved and matured, there has been a growing awareness of the importance of sound industrial hygiene practices and approaches. This can generally be seen in the number of industrial hygienists currently working in the industry, and the emphasis on IH programs in the overall environmental health and safety (EH&S) efforts of the manufacturers in the industry, worldwide. The Semiconductor Industry Association (SIA), through its Occupational Health, Environmental and Fabs Committees, and the Semiconductor Safety Association (SSA) have played key roles in focusing the energies and talent of their members on industrial hygiene issues and concerns.

As sophisticated questions of the general health of workers in the fabrication areas have surfaced, IH monitoring results, hazard evaluations, and participation in the overall occupational health team have been instrumental in performing scientifically sound studies of the industry. Industrial hygiene monitoring information and data were supplied by the member companies of the Semiconductor Industry Association to the University of California's epidemiological study's ("Worker Health Study") three research teams.[1] This IH data has been an integral part of the "exposure assessment" portion of the study.

This book addresses questions of hazard potentials from chemical and physical agents commonly used in semiconductor manufacturing. It is based primarily on results that have been generated from monitoring in some of the larger "merchant" semiconductor manufacturers, either published in the literature or through personal communications with the authors.[2]-[45] These data are intended as a general guide to the maximum potential exposure levels which could be expected to employees doing the types of activities that are listed. These are not intended as hard and fast values that can be used in place of performing actual IH monitoring.

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